

Reetu Paila

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Summary

Final-year Computer Science student seeking a Software Engineer role to design and build scalable, high-availability distributed systems. Strong in C++, Java, and Python with deep foundations in data structures, algorithms, networking, and Linux systems. Experienced in backend development, automation, and performance optimization, building reliable infrastructure and tools that improve development efficiency.

Education

Keshav Memorial Institute Of Technology

Bachelor of Technology in Computer Science | CGPA: 8.3/10

Nov 2022 – Jul 2026

Hyderabad, Telangana

Relevant Coursework

- Data Structures
- Algorithms
- Operating Systems
- Distributed Systems
- Networking & TCP/IP
- Database Management
- Web Development
- Artificial Intelligence

Projects

Network Testbed Builder & Failure Injection System | *Python, Docker, Linux, Prometheus, Grafana*  / Feb 2026

- Built a Docker-based leaf-spine network topology (2 spines, 4 leaves, 8 hosts) with automated deployment and configuration.
- Implemented automated L2, L3, gateway, and spine connectivity tests with PASS/FAIL metrics.
- Designed failure injection for network faults (container shutdown, packet loss, route disruption) and validated recovery.
- Generated HTML reports summarizing baseline, failure, and recovery scenarios for debugging and analysis.
- Integrated Prometheus & Grafana along with Linux networking tools to observe traffic behavior and system performance.

Embedded IoT System | *C, Linux, Bash, System Programming*

 / Dec 2025

- Developed a simulated embedded Linux environment for board bring-up and firmware validation without physical hardware.
- Implemented boot log analysis to detect kernel panics, failures, and expected initialization sequences.
- Built CPU and memory monitoring modules using system interfaces (`/proc/stat`, `/proc/meminfo`).
- Designed automated test workflows using Python and pytest to validate system behavior across multiple scenarios.
- Generated HTML reports summarizing system health, failures, and performance metrics for debugging and analysis.

Relay Chat Platform | *React, Node.js, WebSockets, Express*

 / Nov 2025

- Developed a real-time chat application using WebSockets enabling instant message exchange between multiple clients.
- Built a Node.js backend with event-driven architecture to handle message routing and user connections efficiently.
- Implemented user presence tracking (online/offline) and dynamic message broadcasting across connected clients.
- Designed responsive frontend interface using React for seamless real-time interaction.
- Optimized communication flow to ensure low-latency message delivery and scalability.

Technical Skills

Languages: C, C++, Python, Java, Bash, Go

Linux & Systems: Linux, Embedded Linux, Shell Scripting, Multithreading, Process & Memory Management

Networking: TCP/IP, UDP, VLANs, Routing, Linux Networking Tools, Network Debugging, Troubleshooting

Backend & APIs: Spring Boot, Flask, REST APIs, WebSocket

Automation & Testing: pytest, Python Automation, Regression Testing, Jenkins, CI/CD, Network Validation

Cloud & DevOps: Docker, Kubernetes, Jenkins, Prometheus, Grafana

Databases: MongoDB, PostgreSQL, MySQL

Achievements

- **CCNA 200-301 Certification** 
- Selected among Top 50 finalists in the BITS Pilani National Hackathon for building an AI automation platform within 22 hours.
- Received the Best Project Presentation Award for designing a Face Recognition Attendance System using OpenCV and deep learning.
- Earned the AWS Certified Cloud Practitioner credential validating cloud fundamentals and deployment knowledge.
- Completed multiple Salesforce Trailhead modules, including Admin Beginner, Business Process Automation, Data Management Specialist Superbadge, and App Customization Specialist.